Resume Analysis and Performance Evaluation System

ABSTRACT:-

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1. Introduction

The Resume Analysis and Performance Evaluation System is a software-based application designed to automate the process of resume screening and candidate evaluation. In today's competitive job market, organizations receive a large number of applications for a single job opening, making manual review slow, inefficient, and error-prone. This system provides an intelligent approach to analyze resumes, match them against job requirements, and rank candidates, thereby improving the efficiency and quality of hiring decisions.

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2. Aim of the Project

The main aim of this project is to design and develop a system that can:

- Automate resume reading and information extraction
- Match candidate profiles with job descriptions
- Evaluate and rank candidates based on skills, experience, and qualifications
- Provide a fair, consistent, and unbiased evaluation process

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3. Problem Statement

Traditional recruitment methods rely heavily on manual resume screening, which:

- Consumes significant time and human effort
- inconsistent and prone to errors when handling large volumes of applications
- May introduce personal bias in candidate selection
- Does not provide feedback to candidates for improvement

This project addresses these challenges by automating the resume analysis and evaluation process.

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4. Methodology / Working

The system operates through the following steps:

• The user uploads a resume in PDF or DOCX format
• The system extracts textual content from the uploaded document
• Key information such as skills, education, experience, and certifications is identified
• The extracted data is compared with the given job description
• A scoring algorithm calculates a job-fit score using keyword-based matching techniques
• A skill gap analysis report is generated to highlight missing or weak areas
• Candidates are ranked based on their scores and results are displayed through a user-friendly interface

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5. Technologies Used

• Python (Flask) – Backend framework used for handling application logic and API requests
• HTML, CSS, JavaScript – Frontend technologies used to build the user interface
• MySQL – Database used to store resumes, job descriptions, and evaluation results
• pdfplumber – Library used for extracting text from PDF files
• python-docx – Library used for extractig tnext from DOCX files
• NLTK – Used for text preprocessing (tokenization, stopword removal)
• spaCy – Used for Named Entity Recognition (NER)
• TF-IDF (Scikit-learn) – Used for text vectorization and scoring

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6. Features of the System

• Resume upload and automated text extraction
• Extraction of skills, education, and experience
• Candidate scoring and ranking
• Skill gap identification
• Simple, efficient, and user-friendly interface

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7. Applications / Uses

• Helps recruiters efficiently shortlist candidates
• Reduces time and effort in the hiring process
• Can be used in companies, recruitment agencies, and job portals
• Assists students and job seekers in evaluating and improving their resume

8. Future Scope

- Integration of deep learning models (e.g., BERT) for better semantic matching
- Automated resume improvement suggestions
- Chatbot-based feedback system
- Expansion into a full-scale Applicant Tracking System (ATS)

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9. Conclusion

The Resume Analysis and Performance Evaluation System provides an efficient and scalable solution to modern recruitment challenges. By automating resume screening and candidate evaluation, the system reduces manual effort, minimizes bias, and enhances the overall hiring process, benefiting both recruiters and job applicants.